

NILAVA METYA

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DOB: December 30, 2001 (Age: 23)

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INTRODUCTION

I identify as an applied mathematician. My research encompasses applying techniques from algebra and geometry to algorithmic problems in *optimization* and *machine learning*, especially those that require being robust to uncertain data. More specifically, I like to borrow tools from algebra, convex geometry, Banach space theory and probability.

EDUCATION

Rutgers, the State University of New Jersey - New Brunswick

Doctor of Philosophy in Mathematics | CGPA: 4.0/4.0

(passed written qualifying exams in first attempt just before program started)

Sep '22 – (expected) '27

Piscataway, New Jersey, USA

Chennai Mathematical Institute

Bachelor of Science (Honours) in Mathematics and Computer Science | CGPA: 9.72/10

Position: *Third* (out of 55 students)

(completed degree requirements in 2.5 years)

Aug '19 – May '22

Chennai, Tamil Nadu, India

Don Bosco School, Liluah

Indian School Certificate (ISC) 2019 | Percentage: 97.25%

Position: *First* in science stream (~ 55 students), *second* overall (~ 180 students)

Indian Certificate of Secondary Education (ICSE) 2017 | Percentage: 96.6%

Position: *First* in school (~ 180 students)

Apr '06 – Mar '19

Howrah, West Bengal, India

PUBLICATIONS/PREPRINTS

2. N Metya, A Sinha. Temporal robustness in discrete time linear dynamical systems.
Submitted
1. G DePaul, S Hoşten, N Metya, I Nometa. Degrees of the Wasserstein distance to small toric models.
Journal of Algebraic Statistics

TALKS

INFORMS invited session on AI for Security and Resilience in Complex Networks

Temporal robustness in discrete time linear dynamical systems.

Oct '25

ACMU, Indian Statistical Institute | Host: Sourav Chakraborty

Temporal robustness in discrete time linear dynamical systems.

Aug '25

School of Computing, National University of Singapore | Host: Jonathan Scarlett

Temporal robustness in discrete time linear dynamical systems.

May '25

Mina Rees Women and Math Conference, CUNY

Degrees of the Wasserstein distance to small toric models.

Mar '25

ONGOING PROJECTS

Bilinear robustness in RL (with Arunesh Sinha)

Harmonic analysis and optimization over orbitopes (with Yong Sheng Soh)

Regret bounds in contextual bandits with robustness considerations (with Jonathan Scarlett)

Rank Aggregation problem in the streaming setting (with Prantar Ghosh, Diptarka Chakraborty)

ACADEMIC VISITS

School of Computing, National University of Singapore <i>Prof Jonathan Scarlett</i>	May-Aug '25
Department of Mathematics, Sorbonne University <i>Prof Anne Marie Aubert</i>	May '22

ATTENDANCE IN CONFERENCES/WORKSHOPS

Princeton Machine Learning Theory Summer School <i>Summer School</i> Princeton	Aug '24
Efficient Algorithms for High Dimensional Metrics <i>Workshop</i> DIMACS, Rutgers, New Brunswick	May '24
Bayesian Statistics and Statistical Learning <i>Workshop</i> IMSI, Chicago	Dec '23
Algebraic Statistics for Ecological and Biological Systems <i>Workshop</i> IMSI, Chicago	Oct '23
Apprenticeship Week: Varieties from Statistics IMSI, Chicago	Oct '23
Invitation to Algebraic Statistics and Applications IMSI, Chicago	Sep '23
Algebraic Methods in Biochemical Reaction Networks MPI, Leipzig	Jun '23
Joint Mathematics Meetings Boston	Jan '23
AlGeCom-XII (Algebra Geometry and Combinatorics day) UIUC	Oct '22

SELECTED COURSEWORK

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|---------------------------|---|---|
| • Quantum Computation | • Haskell | • Topological Data Analysis ^{Rutgers} |
| • Matrix Computations | • Algorithm Design + Analysis | • Convex Optimization ^{Princeton} |
| • Statistics with R | • Algebraic Geometry ^{Princeton} | • Convexities in Banach Spaces ^{Princeton} |
| • Algebraic Topology | • Homological Algebra ^{Rutgers} | • Random Matrices ^{Princeton} |
| • Complex Analysis | • Probability ^{Rutgers} | |
| • Quiver Representations | • Data Mining ^{Rutgers} | |
| • Algebraic Number Theory | | |

TEACHING AND GRADING

Workshop leader for Calculus II Rutgers	Sep – Dec '23, Jan – Apr '24
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Head Counselor at PROMYS India IISc Bangalore	May – Jun '23
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Grader | Rutgers University

Linear Algebra and Applications	Jan – Apr '25
Mathematical Statistics	Sep – Dec '24
Algebra II	Jan – Apr '24
Linear Algebra and Applications	Sep – Dec '23
Analysis II	Jan – Apr '23
Topics in Applied Algebra	Jan – Apr '23
Topology	Sep – Dec '22
Theory of Numbers	Sep – Dec '22

Teaching Assistant | Chennai Mathematical Institute

Algebra II (Group theory)	BSc 1st year	<i>Prof Manoj Kummini</i>	Jan – May '22
Algebra I (Linear algebra)	BSc 1st year - head tutor	<i>Prof T R Ramadas</i>	Sep – Dec '21
Functional Programming in Haskell	BSc and MSc Comp. Sci. 1st year	<i>Prof S P Suresh</i>	Sep – Dec '21
Probability Theory	BSc 1st year	<i>Prof P Sankaran</i>	Apr – Jul '21
Discrete Mathematics	BSc 1st year	<i>Prof K V Subrahmanyam</i>	Apr – Jul '21
Design and Analysis of Algorithms	MSc Data Science 1st year	<i>Prof G Philip</i>	Apr – Jul '21
Algebra I (Linear algebra)	BSc 1st year	<i>Prof T R Ramadas</i>	Dec '20 – Mar '21
Functional Programming in Haskell	BSc and MSc Comp. Sci. 1st year	<i>Prof S P Suresh</i>	Dec '20 – Mar '21

Counselor at PROMYS Boston University	Jul – Aug '20, '21
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HONOURS AND AWARDS

Nominated by Rutgers Math department for SLMath summer school **Jun '23**
Summer school at Leipzig - awarded full travel funding. Only two students from Rutgers Math were fully funded by SLMath.

Academic Excellence Award at Rutgers **Sep '22**
Received a certificate and \$100 based on performance in Written Qualifying Exams.

Shriram Scholarship at CMI **'19 – '22**
Received institutional fee waiver and monthly stipend (based on entrance exam).

Ranked 4th nationally at the Bachelor of Statistics (B.Stat.) entrance examination **'19**
Indian Statistical Institute (ISI)

Informatics Olympiad **'17, '18, '19**
Selected among (approx) top 100-130 school students in India in Zonal Informatics Olympiad (ZIO).

Mathematical Olympiad **Jan '18**
Selected for Indian National Mathematical Olympiad (INMO) Training Camp | top 30 school students in West Bengal.

Program in Mathematics for Young Scientists (PROMYS) **'18, '19, '20, '21**
Awarded the Tara and Jasubhai Mehta Fellowship to PROMYS (among 5 Indian school students in 2018) based on a competitive process. Participated twice as a student ('18, '19) and twice as a counselor ('20, '21).

Others

- Qualified for **International Collegiate Programming Contest (ICPC)** Kharagpur regionals and Amritapuri regionals in 2019 and secured rank 35 among (approx) 90 university teams at Kharagpur.
- Selected among top 30 students in India to participate in **Scholastic Test of Excellence in Mathematical Sciences (STEMS)** camp at CMI in 2018, based on a competitive exam (across grades 9 – 12 and across Math, Physics, Computer Science).
- Secured the **third position** in **Mathematics Talent Reward Programme (MTRP)** 2016, organized by ISI Kolkata, based on a competitive exam and quizzes at a camp.

SERVICE

Coffee Hour (departmental socializing) | *Coordinator* | Rutgers Math Department **Jan '25 -**
Physics of Learning Theory (PhyLT) | *Organizer* | Rutgers Math Department **Jan '25 -**
Written Qualifying Exams: Algebra | *Instructor* | Rutgers Math Department **Jun - Aug '23**
Algebra 'N' Geometry Learning Seminar (ANGeLS) | *Organizer* | Rutgers Math Department **Jan - Apr '23**
Student Seminar | *Organizer* | Chennai Mathematical Institute **Oct - Dec '22**
ICO Camp (online) | *Combinatorics teacher* | CodeChef **Nov '20**

SKILLS

Languages	Bengali (mother tongue), English (fluent), Hindi (fluent)
Programming	JAVA, C++, Python, Haskell, R, HTML, SageMath, Macaulay2, MATLAB
Documentation	L ^A T _E X, Microsoft Word